

MULTI-AGENT URBAN TRAFFIC SIGNAL OPTIMIZATION IN SIMULATED ENVIRONMENTS

Project Id: 25-26J-523

Project Proposal Report

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Engineering

Department of Computer Science and Software Engineering

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Date: 20-08-25

ABSTRACT

This research proposes the development of an advanced, AI-driven optimizer for multi-intersection traffic signal control to address the critical challenges of urban traffic congestion. The central research problem is that existing Graph Neural Network (GNN) based solutions are limited by oversimplified network representations, a lack of safety-aware mechanisms, and inefficient learning paradigms. The basic design of this study is a novel GNN component, architected as a recurrent Heterogeneous Graph Attention Network (HGAT), which operates on a dynamic and heterogeneous graph to more realistically model complex urban environments. Key novelties include the integration of principled uncertainty quantification using Monte Carlo (MC) Dropout for robust, safety-aware decision-making, and a hybrid training strategy that combines Self-Supervised Learning (SSL) for pre-training with Multi-Agent Reinforcement Learning (MARL) for fine-tuning. The component is designed as part of a modular, verifiable system, complete with a real-time dashboard and manual control interface, all operating within a SUMO simulation environment. It is anticipated that this approach will result in a controller that significantly outperforms traditional and baseline AI models in key traffic efficiency metrics, such as average vehicle travel time and queue length. The project will deliver a robust proof-of-concept for a next-generation adaptive traffic management system, demonstrating a clear pathway from a theoretical algorithm to a practical, deployable solution.

Keywords—Traffic Signal Control, Graph Neural Networks (GNNs), Multi-Agent Reinforcement Learning (MARL), Uncertainty Quantification, Heterogeneous Graphs, Self-Supervised Learning (SSL), Spatio-Temporal Modeling..

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LIST OF ABBRIVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
CTDE	Centralized Training with Decentralized Execution
DQN	Deep Q-Network
GAT	Graph Attention Network
GCN	Graph Convolutional Network
GNN	Graph Neural Network
GPU	Graphics Processing Unit
GRU	Gated Recurrent Unit
HGAT	Heterogeneous Graph Attention Network
ITS	Intelligent Transportation Systems
KPI	Key Performance Indicator
MARL	Multi-Agent Reinforcement Learning
MC Dropout	Monte Carlo Dropout
MPC	Model Predictive Controller
REST	Representational State Transfer
RL	Reinforcement Learning
SSL	Self-Supervised Learning
SUMO	Simulation of Urban MObility
TCN	Temporal Convolutional Network
TraCI	Traffic Control Interface
UML	Unified Modeling Language

1. INTRODUCTION

The effective management of urban traffic is a cornerstone of modern city planning and a critical component of Intelligent Transportation Systems (ITS). Escalating traffic congestion impedes mobility, carries significant economic costs, and contributes to environmental degradation. While traditional statistical and machine learning methods have been applied to this problem, their effectiveness is limited by the complex, dynamic, and non-Euclidean nature of road networks [1]. The advent of deep learning, particularly Graph Neural Networks (GNNs), has marked a significant paradigm shift in the field. GNNs are uniquely suited to model the inherent graph structure of transportation networks, enabling them to learn and reason about the complex spatial and relational dependencies between intersections [1].

To develop this project, a mastery of several key techniques that represent the current state-of-the-art is essential. Foundational studies in traffic forecasting have established the effectiveness of combining GNNs with sequential models to capture both spatial and temporal dependencies simultaneously. For instance, the Temporal Graph Convolutional Network (T-GCN) integrates a Graph Convolutional Network (GCN) for spatial feature extraction with a Gated Recurrent Unit (GRU) to model traffic changes over time [2]. This hybrid approach has been shown to outperform simpler deep learning models by explicitly modeling the underlying network structure. Building on this, the field has advanced toward active, adaptive traffic signal control. This is largely achieved by integrating GNNs with Multi-Agent Reinforcement Learning (MARL), where each intersection is treated as an intelligent agent. Within this framework, advanced spatial modules like the Graph Attention Network (GAT) are employed to allow agents to weigh the importance of information from neighboring intersections, a crucial capability for effective coordination [6].

The current state of the art is characterized by a move away from static, predefined graphs towards dynamic structures that better reflect the evolving state of a live traffic network. Previous work has explored this in two primary ways. Some models, like the Progressive

Graph Convolutional Network (PGCN), learn an adaptive adjacency matrix that dynamically constructs graph connections based on the trend similarity of online data during both training and testing [4]. Other approaches, such as the Spatio-temporal Causal Graph Attention Network (STCGAT), use node adaptive learning to generate new graph structures at each time step [5]. However, a comprehensive analysis of the literature reveals several limitations in existing work. Firstly, the majority of current models, including T-GCN, STGCN, and PGCN, treat the traffic network as a homogeneous graph, failing to capture the diverse nature of a real urban environment which contains distinct entities like pedestrian crossings and bus lanes [2], [3], [4]. While the HGAT-MARL paper highlights the importance of heterogeneous modeling, its focus is primarily on the control aspect rather than a comprehensive GNN architecture [6]. Secondly, the literature largely overlooks the need for the model to quantify its own confidence, providing deterministic outputs without a measure of reliability, which is a significant drawback for a safety-critical system. Finally, training MARL agents is notoriously sample-inefficient, and there is less focus on using advanced techniques like Self-Supervised Learning (SSL) to accelerate training and develop more robust policies [1].

Our approach builds directly upon these pioneering works while aiming to address their core limitations. We adopt the successful spatio-temporal architecture and the MARL framework as our foundation, leveraging the promising concepts of heterogeneous graphs from HGAT-MARL [6] and dynamic graph construction from PGCN [4]. Our work varies from and extends the state-of-the-art by introducing four key novelties. First, we propose a GNN controller built upon a graph that is both heterogeneous and dynamic, offering a more realistic and powerful representation of the urban traffic environment. Second, we incorporate principled uncertainty quantification using Monte Carlo (MC) Dropout, a critical feature for safety and robustness that is currently missing in traffic control models. Third, we implement a hybrid training strategy, using Self-Supervised Learning (SSL) for pre-training followed by MARL fine-tuning to improve learning efficiency. Finally, our component is designed as a modular and interactive system, complete with a real-time

dashboard and manual control interface, addressing the gap between theoretical algorithms and practical, verifiable deployment.

1.1 Background and Literature Review

The field of Intelligent Transportation Systems (ITS) has undergone a significant transformation with the advent of data-driven methodologies. Historically, traffic forecasting and control relied on statistical models like ARIMA or traditional machine learning techniques, which struggled to capture the complex, non-linear dynamics inherent in urban traffic networks [1]. The recognition of road networks as inherent graph structures has propelled Graph Neural Networks (GNNs) to the forefront of ITS research, providing a powerful framework for modeling these complex systems.

Recent literature overwhelmingly supports GNNs as the state-of-the-art approach for traffic applications due to their inherent ability to model the complex, non-Euclidean nature of road networks. Traditional deep learning models like Convolutional Neural Networks (CNNs), which are designed for grid-like data such as images, struggle to effectively capture the irregular structure of road systems. As noted in a comprehensive survey by Jiang and Luo, GNNs are "ideally suited" for this domain because they directly process the graph structure of a road network—with intersections as nodes and roads as edges—allowing them to capture spatial dependencies that other models cannot [7]. This native ability to integrate both the network's topology and its dynamic data consistently leads to more robust modeling. In contrast, "CNNs miss directionality and context; GNNs integrate topological and dynamic data," which is crucial for the complex interactions in traffic control [8]. This has cemented the consensus that GNNs have "emerged as state-of-the-art traffic forecasting solutions because they are well suited for traffic systems with graph structures" [9].

The core challenge in traffic analysis is to effectively model dependencies that exist across both space and time. State-of-the-art research has converged on hybrid architectures that combine dedicated spatial and temporal modules to address this.

The spatial module is responsible for capturing the influence of the road network's topology. Early and effective models like the Temporal Graph Convolutional Network (T-GCN) utilized a Graph Convolutional Network (GCN), which operates by aggregating feature information from a node's immediate neighbors [2]. While powerful, a standard GCN is isotropic, meaning the weights used for aggregation are fixed and based on the graph's structure (e.g., node degree), treating all neighbors with equal importance. To overcome this limitation, recent work has shifted towards Graph Attention Networks (GATs). GATs introduce a self-attention mechanism that dynamically computes the importance of each neighboring node in a given context, allowing the model to learn an anisotropic weighting scheme [5]. This is particularly relevant in traffic networks, where an upstream intersection on a major artery might have a greater influence than one on a quiet side street. Models like STCGAT and HGAT-MARL leverage this attention mechanism to achieve a more nuanced understanding of spatial relationships [5, 6].

The temporal module captures the dynamic evolution of traffic patterns over time. The most common approach involves using Recurrent Neural Networks (RNNs) or their advanced variants, such as the Gated Recurrent Unit (GRU) or Long Short-Term Memory (LSTM) networks. The T-GCN model, for example, feeds the spatial embeddings from its GCN into a GRU to process the time-series data sequentially [2]. An alternative and computationally efficient approach is the use of Temporal Convolutional Networks (TCNs) or dilated causal convolutions, as seen in the Progressive Graph Convolutional Network (PGCN), which can process long sequences in parallel and exponentially increase their receptive field without requiring an excessively deep architecture [4].

A key limitation of early GNN models was their reliance on a single, static adjacency matrix to represent the road network. This approach fails to capture the fact that the

functional relationships between intersections can change dramatically based on real-time traffic conditions.

To address this, research has moved towards dynamic graph structures. One line of work focuses on learning an adaptive graph during the training phase, where node embeddings are used to generate a static, data-driven adjacency matrix that captures latent correlations beyond simple physical connectivity [4]. However, a more advanced approach is online or progressive adaptation, where the graph itself is updated at each time step during both training and testing. The PGCN model achieves this by constructing a "progressive graph" based on the parameterized cosine similarity of node signal trends, making it highly responsive to immediate traffic changes [4]. Similarly, STCGAT uses a "node adaptive learning" mechanism to generate a dynamic subgraph at each time step [5].

Another critical advancement is the move from homogeneous to heterogeneous graphs. Most models, including T-GCN and PGCN, treat all elements of the network as a single type of node (intersections/sensors) and edge (roads) [2, 4]. This is a significant oversimplification of a real urban environment. The HGAT-MARL paper demonstrates a more sophisticated approach by modeling the traffic network as a heterogeneous graph with distinct node types for Signal Controllers, Connections, Lanes, and Vehicles [6]. This allows the model to learn type-specific representations and interaction rules, providing a much richer and more realistic foundation for decision-making.

While traffic forecasting is a mature application area, the ultimate goal is active control. The dominant paradigm for coordinating multiple intersections is Multi-Agent Reinforcement Learning (MARL) [1]. In this framework, each intersection is controlled by an independent RL agent that learns a policy to optimize a local or global objective.

The integration of GNNs into MARL has proven to be a powerful combination. The GNN serves as a sophisticated state encoder, providing each agent with a rich, contextual understanding of its local neighborhood by aggregating information from adjacent intersections. This allows agents to make decisions that are not just locally optimal but

also coordinated with their neighbors. The HGAT-MARL model exemplifies this, using a Heterogeneous GAT to process the complex state of the traffic network before feeding it into a Deep Q-Network (DQN) for each agent [6]. A common and effective training strategy in this domain is Centralized Training with Decentralized Execution (CTDE). During training, a central controller has access to the global state and actions of all agents, allowing it to efficiently train a coordinated policy. During execution, each agent acts independently using only its local observations, making the system scalable and practical for real-world deployment [6].

1.2 Research Gap

The existing literature demonstrates significant progress in applying Graph Neural Networks (GNNs) to transportation problems, particularly in moving from traffic forecasting to adaptive traffic signal control [1]. State-of-the-art models effectively combine spatial modules like GCNs or GATs with temporal modules like GRUs or TCNs to analyze spatio-temporal data [2, 5]. Furthermore, a clear trend is the shift from static graph representations to dynamic graphs that can adapt to evolving traffic conditions, as seen in models like PGCN [4]. However, a thorough review of these advanced methods reveals several critical gaps in the existing research literature, which this project aims to address.

A primary limitation is the lack of heterogeneity in traffic object representation. The majority of current models, even sophisticated ones like T-GCN, STGCN, and PGCN, treat the traffic network as a homogeneous graph, where all nodes (e.g., intersections) and edges (e.g., roads) are of a single type [2, 3, 4]. This is an oversimplification that fails to capture the diverse nature of a real urban environment, which contains distinct entities such as standard intersections, pedestrian crossings, bus lanes, and various vehicle types. While the HGAT-MARL paper introduces a heterogeneous graph, its focus is primarily on the Multi-Agent Reinforcement Learning (MARL) control aspect rather than on

developing a comprehensive and modular GNN architecture that fully leverages this rich representation [6].

Secondly, there is a notable absence of principled uncertainty quantification for control. The reviewed literature shows that GNNs are increasingly used for high-stakes decision-making in traffic signal control. However, these models provide deterministic outputs without an associated measure of their own confidence or reliability. This is a significant drawback for a safety-critical system, as the model cannot signal when it is encountering an "out-of-distribution" event (like an unforeseen accident or sensor failure) where its decisions may be unreliable [1].

Thirdly, existing approaches often suffer from inefficient learning paradigms. Training a MARL system from scratch is notoriously sample-inefficient and time-consuming. While some works use imitation learning, there is a clear gap in the systematic application of Self-Supervised Learning (SSL) as a preliminary phase. SSL could allow the GNN to learn the fundamental "physics" of traffic flow from vast amounts of unlabeled simulation data *before* the complex RL fine-tuning begins, thereby accelerating training and leading to a more robust final policy [1].

Finally, there is a gap between the development of novel algorithms and their integration into practical, verifiable systems. Many research papers focus purely on the algorithmic contribution and its performance metrics, often lacking a clear, modular software engineering blueprint that includes crucial components like real-time monitoring dashboards, manual override capabilities, and public-facing APIs for data dissemination. The following table provides a direct comparison of the key features of existing research against our proposed solution, visually highlighting the gaps this project will fill.

Table 1 Comparison of Existing Research

Features	T-GCN [2]	STGCN [3]	PGCN [4]	HGAT-MARL [6]	Our Proposed Solution
Primary Task: Traffic Signal Control	✗	✗	✗	✓	✓
Heterogeneous Graph Representation	✗	✗	✗	✓	✓
Dynamic Graph Adaptation (Online)	✗	✗	✓	✗	✓
Principled Uncertainty Quantification	✗	✗	✗	✗	✓
Self-Supervised Pre-training	✗	✗	✗	✗	✓
Integrated System with Dashboard	✗	✗	✗	✗	✓

This proposal seeks to fill these identified gaps by developing a holistic GNN-based traffic signal optimizer. Our approach will achieve this by fusing a heterogeneous and dynamic graph structure for a more realistic traffic representation, incorporating principled uncertainty quantification for safe and robust decision-making, implementing a hybrid SSL/RL training paradigm for efficient learning, and designing a complete and interactive system architecture for practical validation and deployment.

1.3 Research Problem

In the domain of urban traffic management, existing intelligent transportation systems (ITS) still struggle to achieve optimal, network-wide signal control, leading to persistent congestion, travel delays, and increased fuel consumption. While the application of Graph Neural Networks (GNNs) has shown significant promise, current state-of-the-art models often rely on oversimplified assumptions about the traffic environment and lack critical features for safe and practical deployment [1]. This leads to a central research problem that this project aims to address:

How can a GNN-based system be designed to overcome the limitations of existing models by integrating a more realistic and dynamic traffic network representation with principles of safe AI, efficient learning, and practical system design to achieve superior, network-wide traffic signal optimization?

To solve this overarching problem, this research must address several specific, interconnected issues that current methodologies do not adequately resolve:

- **The Representation Problem:** How can the diverse and dynamic nature of urban traffic—including different types of road users (vehicles, buses, pedestrians) and infrastructure (intersections, bus lanes)—be modeled more effectively than with current homogeneous, static graph structures? Existing models like T-GCN and PGCN treat all network elements as a single type, which fails to capture the unique behaviors and interactions of these heterogeneous entities [2, 4].
- **The Safety and Reliability Problem:** How can a GNN-based control agent be made aware of its own predictive limitations to ensure robust and safe operation, especially during novel or unexpected traffic events such as accidents or sensor failures? The current literature lacks a focus on principled uncertainty quantification, meaning models make critical decisions without communicating their level of confidence, a significant risk in a safety-critical application [1, 6].
- **The Learning Efficiency Problem:** How can the notoriously sample-inefficient and time-consuming training process of Multi-Agent Reinforcement Learning (MARL) be accelerated to produce a high-performing and generalizable control policy more effectively? The "cold start" problem remains a significant hurdle, and there is a research gap in systematically applying advanced techniques like Self-Supervised Learning (SSL) to pre-train models on fundamental traffic dynamics before control-specific fine-tuning [1].
- **The Practicality and Verification Problem:** How can a complex AI algorithm be developed not in isolation, but as a component within an integrated and

verifiable system? The majority of research focuses on algorithmic performance, neglecting the design of essential tools for real-world deployment, such as live monitoring dashboards and manual override capabilities that are crucial for debugging, validation, and building trust with human operators [3].

This research directly confronts these challenges by proposing the design, implementation, and evaluation of a novel GNN-based optimizer that holistically integrates solutions to each of these problems.

2. OBJECTIVES

2.1 Main Objectives

The main objective of this component is to design, implement, and evaluate a novel, GNN-based traffic signal optimizer that demonstrates a measurable improvement in network-wide traffic flow within a simulated urban environment. This will be achieved by developing a system that integrates a heterogeneous and dynamic graph representation with principled uncertainty quantification and an efficient hybrid training strategy, culminating in a functional software component by the end of the academic year.

2.2 Specific Objectives

To achieve the main objective, the following specific, time-bound objectives will be met:

- **To develop a robust data pipeline and dynamic graph construction module by the end of the initial implementation phase.** This involves creating a Python-based module that interfaces with the SUMO simulator via the TraCI API, parses the road network into a dynamic and heterogeneous graph structure, and successfully ingests real-time traffic data (e.g., queue lengths, flow rates) at each simulation tick. **(Addresses the Representation Problem)**
- **To implement and validate a recurrent Heterogeneous Graph Attention Network (HGAT) with integrated uncertainty quantification by the midpoint of the development phase.** This objective entails building the core GNN model in PyTorch Geometric and integrating Monte Carlo (MC) Dropout. The success of this objective will be measured by the model's ability to output both a valid control action and a quantifiable epistemic uncertainty score for each intersection. **(Addresses the Representation Problem and the Safety and Reliability Problem)**
- **To develop and execute a two-phase hybrid training pipeline by the end of the development phase.** This involves implementing a Self-Supervised Learning

(SSL) framework for pre-training the GNN on unlabeled simulation data, followed by fine-tuning the model as a Multi-Agent Reinforcement Learning (MARL) agent. The objective's success will be determined by the convergence of the SSL loss and a clear positive trend in the MARL agent's cumulative reward during training. **(Addresses the Learning Efficiency Problem)**

- **To implement a real-time monitoring dashboard with a manual control interface in time for the final system testing phase.** This requires building a decoupled web application (Node.js and React) that visualizes key performance indicators (KPIs), GNN decisions, and model uncertainty from the live simulation. A key deliverable is a functional manual override feature that allows a human operator to take control of a specific traffic signal. **(Addresses the Practicality and Verification Problem)**
- **To design and deploy a public-facing API for traffic data dissemination for the final project submission.** This involves creating and documenting a RESTful API with functional endpoints that allow external users to query real-time and forecasted traffic data (e.g., delays, congestion levels) from the system. **(Addresses the Practicality and Verification Problem)**

3. METHODOLOGY

This section details the comprehensive plan for developing and evaluating the GNN-Based Traffic Signal Optimizer. It outlines the system architecture, the step-by-step implementation plan, the resources required, data collection procedures, a project timeline, and the anticipated outcomes of the research.

System Architecture

This project aims to create a modular and intelligent traffic signal control system by integrating advanced AI models within a robust and interactive software architecture. The purpose of the project is to provide traffic engineers and city planners (admins) with an intuitive web-based platform for real-time monitoring and management of urban traffic flow. This user interface will support several key functions, from visualizing network-wide congestion patterns to analyzing the performance of the AI controller and allowing for manual intervention at specific intersections.

The system is designed to seamlessly integrate technological solutions with sound software engineering principles to create a cohesive and powerful experience for all users, with the resultant outcome being improved traffic efficiency and reduced congestion.

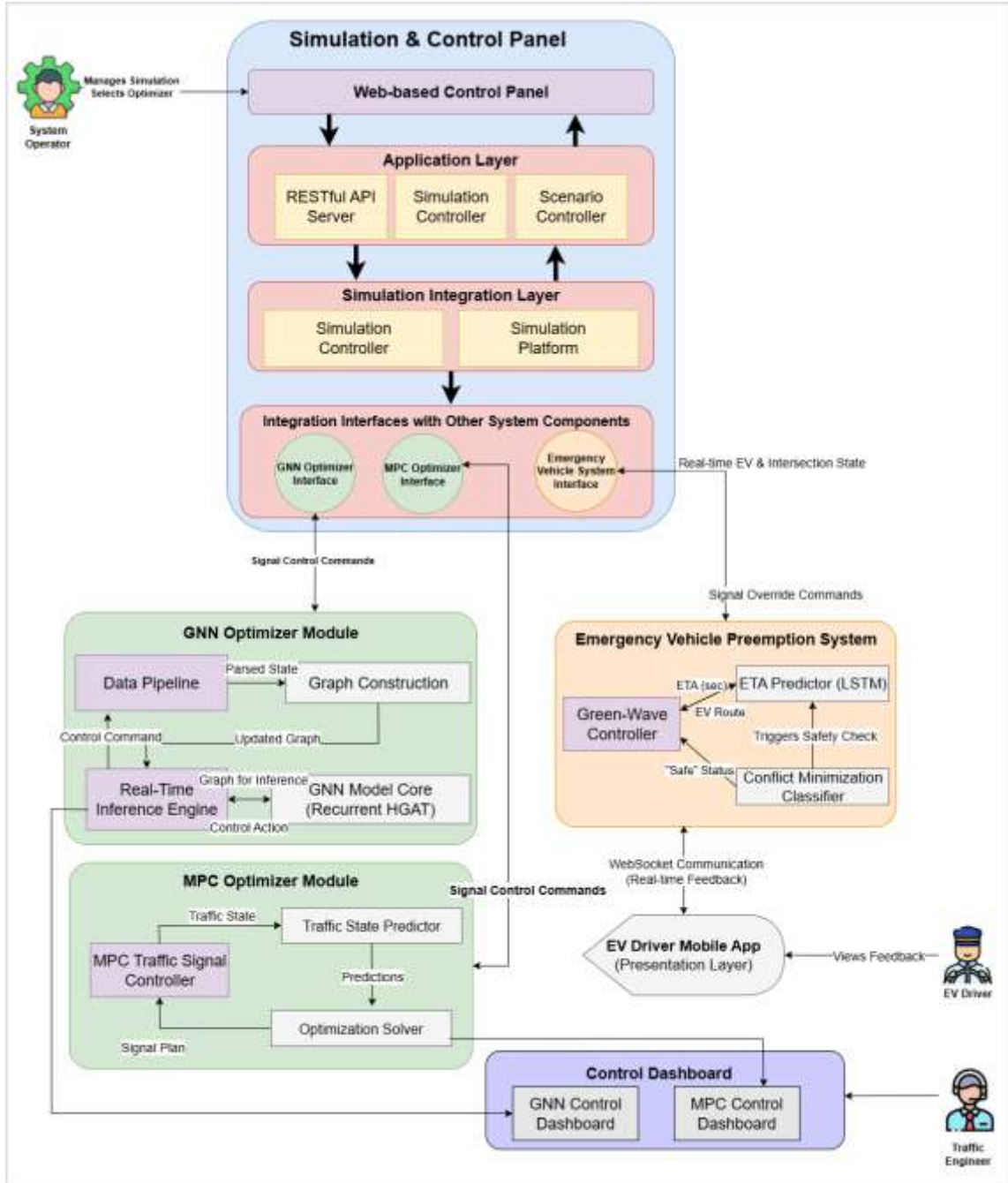


Figure 3.1 System diagram for overall system

The following components, illustrated in the system architecture diagram below, describe the full workflow of the GNN-Based Traffic Signal Optimizer. The explanation details

how each block of the pipeline works together to provide a seamless solution for traffic engineers to manage and optimize urban traffic.

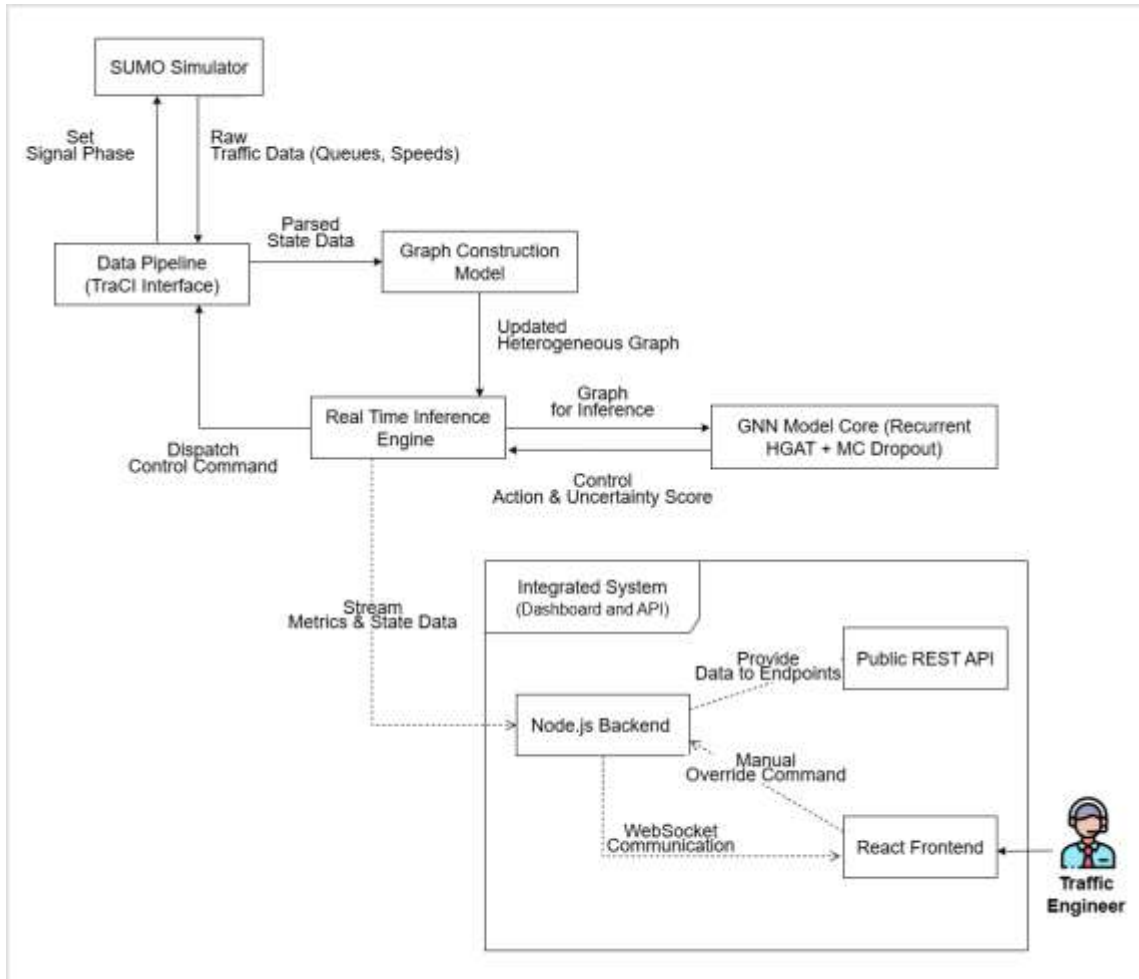


Figure 3.2 System architecture diagram for the component

The system operates around a client-server architecture. The primary interaction for an admin user (e.g., a traffic engineer) occurs through a React-based frontend. This web application serves as the main command center, offering a user-friendly interface for visualizing real-time traffic data, monitoring the AI's decisions, and manually controlling signals.

1. Real-time Data Visualization and Monitoring:

The frontend provides a live, intuitive view of the entire traffic network. Admins can observe key performance indicators such as average vehicle delay, network-wide queue lengths, and throughput. A central feature is a map-based visualization where intersections are color-coded based on congestion levels or the AI model's decision uncertainty, allowing for quick identification of problem areas.

2. Manual Traffic Control:

For situations requiring human intervention, the dashboard allows an admin to select any intersection and manually override the AI's control. They can force a specific signal phase to address unforeseen events like accidents or special traffic conditions. This feature is crucial for building trust in the system and ensuring operational stability.

3. Public User Interaction:

For public users, the system exposes a RESTful API that allows them to query real-time and forecasted traffic data. A lightweight mobile interface could consume this API to provide users with information on current travel times, congestion levels, and predicted delays on their routes.

Software Architecture and Data Flow

The backend of the system is architected as a set of decoupled services to ensure scalability and maintainability. At the heart of the system is the Python-based GNN controller, which continuously interacts with the SUMO traffic simulator via the TraCI API.

- **GNN Controller:** This core component ingests real-time traffic data (queue lengths, speeds) from SUMO. It uses the Graph Construction Module to update a dynamic, heterogeneous graph representation of the network. This graph is then processed by a recurrent Heterogeneous Graph Attention Network (HGAT), which determines the optimal signal timing for each intersection. The controller then dispatches these commands back to SUMO.

- Backend Server (Node.js): The Python controller streams metrics and state data asynchronously to a Node.js server. This backend server manages a WebSocket connection with the React frontend, pushing live data to the admin dashboard for visualization. It also handles incoming manual override commands and hosts the public-facing RESTful API.

A central database is used to log historical performance data, model decisions, and system events. This allows for long-term analysis of the AI's effectiveness and provides the data needed for retraining or fine-tuning the GNN model in the future.

Overall, the diagram shows how these components work in concert to provide a powerful, user-friendly platform for the detailed analysis and optimization of urban traffic. This system, through a combination of state-of-the-art GNN techniques and a responsive, interactive user interface, offers a robust tool for all stakeholders in the effective management of city-wide traffic flow.

4. PROJECT REQUIREMENTS

This section details the comprehensive requirements for the GNN-Based Traffic Signal Optimizer component. It covers the needs of the end-users, the specific functionalities the system must perform, its quality attributes, and the technical environment required for its operation and development.

4.1 User Requirements

The system is designed for two primary types of users: Admin Users (Traffic Engineers/Researchers) and Public Users.

Admin User:

1. The user must be able to log in to a secure web-based dashboard.
2. The user must be able to view a real-time, map-based visualization of the entire traffic network.
3. The user must be able to monitor key performance indicators (KPIs) such as average travel time, network-wide queue lengths, and system throughput.
4. The user must be able to select an individual intersection to view its specific real-time data and the AI's current control decision and uncertainty level.
5. The user must have the ability to manually override the AI controller for any selected intersection and set a specific signal phase.

Public User:

1. The user should be able to access real-time and forecasted traffic information (e.g., travel times, congestion levels) via a public-facing API or a simple mobile interface.

4.2 System Requirements

- **Development Environment:**
 - A high-performance computer with a dedicated NVIDIA GPU (e.g., RTX 30-series or higher) with at least 8GB of VRAM.
 - Python 3.8+, Node.js, and a modern web browser.
- **Deployment Environment (Tentative):**
 - Cloud-based virtual machines for hosting the Python controller and Node.js backend.
 - A NoSQL database service (e.g., MongoDB Atlas) for data logging.
 - A web server for the React frontend.

4.3 Functional Requirements

- **FR1: Data Ingestion:** The system shall establish a persistent connection to the SUMO simulation environment using the TraCI API. It must be capable of subscribing to and ingesting real-time traffic data—including vehicle counts, speeds, and queue lengths—from every relevant sensor and lane in the network at each one-second simulation step.
- **FR2: Dynamic Graph Construction:** The system shall take the raw, ingested data and transform it into a dynamic, heterogeneous graph representation in real-time. This involves updating the feature vectors of all nodes (intersections, etc.) and edges (roads) at each time step to reflect the current state of the network.
- **FR3: AI-based Signal Control:** The core function of the system is to process the dynamically constructed graph through its GNN model and produce a valid signal phase decision for every controlled intersection in the network. This

decision-making process must be completed within the real-time constraints of the simulation.

- **FR4: Uncertainty Calculation:** For each decision it makes, the system's GNN shall calculate and output a quantifiable epistemic uncertainty score. This score will represent the model's confidence in its own decision, providing a crucial metric for evaluating the reliability of the AI's control actions in different scenarios.
- **FR5: Real-time Visualization:** The system shall display the live state of the traffic network and the outputs of the AI model on the admin dashboard. This includes visualizing KPIs, mapping congestion levels, and showing the current signal phase and uncertainty score for each intersection.
- **FR6: Manual Override:** The system shall provide an interface on the dashboard for an admin user to manually select an intersection and a signal phase. The system must prioritize this user command, overriding the AI's decision for that specific intersection for the current control step.
- **FR7: Public API:** The system shall expose a secure and documented RESTful API. This API will have specific endpoints (e.g., /api/travel-time, /api/congestion) that allow authenticated public users or applications to query a sanitized subset of the system's real-time and forecasted traffic data.

4.4 Non-Functional Requirements

1. **NFR1: Performance:** The system must adhere to a strict real-time performance constraint. The entire inference cycle—from data collection to command dispatch—must be completed in under one second. This is a hard deadline required to maintain synchronization with the simulator and ensure stable control.

2. **NFR2: Scalability:** The software architecture shall be designed in a modular fashion, with clear separation of concerns between the data pipeline, the AI model, and the dashboard services. This will allow the system to be scaled up to larger road networks in the future with minimal refactoring.
3. **NFR3: Reliability:** The system must be able to run for extended simulation periods (e.g., a full 24-hour cycle) without crashing or requiring a restart. All services must include robust error handling and logging to ensure high availability.
4. **NFR4: Usability:** The admin dashboard interface shall be designed to be highly intuitive and user-friendly. A traffic engineer with minimal training should be able to easily understand the visualizations, assess the network's performance, and utilize the system's features effectively.
5. **NFR5: Security:** All access to the admin dashboard and control system shall be protected by a secure authentication and authorization mechanism to prevent unauthorized access and manipulation of the traffic control system.

1. REFERENCES

- [1] H. Li, Y. Zhao, Z. Mao, Y. Qin, Z. Xiao, J. Feng, Y. Gu, W. Ju, X. Luo, and M. Zhang, “Graph Neural Networks in Intelligent Transportation Systems: Advances, Applications and Trends,” *JOURNAL OF LATEX CLASS FILES*, vol. 14, no. 8, pp. 1-20, Aug. 2021. [Online]. Available: arXiv:2401.00713v3
- [2] L. Zhao, Y. Song, C. Zhang, Y. Liu, P. Wang, T. Lin, M. Deng, and H. Li, “Temporal Graph Convolutional Network for Urban Traffic Flow Prediction Method,” *JOURNAL OF LATEX CLASS FILES*, vol. 14, no. 8, pp. 1-10, Aug. 2015. [Online]. Available: arXiv:1811.05320v2
- [3] S. Komarovskiy and J. Haddad, “Spatio-temporal Graph Convolutional Neural Network for traffic signal prediction in large-scale urban networks,” *Transportation Research Interdisciplinary Perspectives*, vol. 32, p. 101482, 2025.
- [4] Y. Shin and Y. Yoon, “PGCN: Progressive Graph Convolutional Networks for Spatial-Temporal Traffic Forecasting,” *IEEE Transactions on Intelligent Transportation Systems*, pp. 1-12, 2024.
- [5] W. Zhao, S. Zhang, B. Zhou, and B. Wang, “STCGAT: A Spatio-temporal Causal Graph Attention Network for traffic flow prediction in Intelligent Transportation Systems,” *JOURNAL OF LATEX CLASS FILES*, vol. 14, no. 8, pp. 1-9, Aug. 2021. [Online]. Available: arXiv:2203.10749v3
- [6] Z. Zhai, R. Hao, B. Cui, and S. Wang, “HGAT and Multi-Agent RL-Based Method for Multi-Intersection Traffic Signal Control,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 5, pp. 6848-6864, May 2025.
- [7] W. Jiang and J. Luo, “Graph Neural Network for Traffic Forecasting: A Survey,” arXiv preprint arXiv:2101.11174, 2022.
- [8] L. Xie et al., “Probabilistic Graph Neural Networks for Traffic Signal Control,” *IEEE Transactions on Intelligent Transportation Systems*, Jun. 2021, doi: 10.1109/TITS.2021.3084840.
- [9] Z. Wu et al., “Graph Neural Network for Traffic Forecasting: The Research Progress,” *ISPRS International Journal of Geo-Information*, vol. 12, no. 3, p. 100, Mar. 2023.